

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12398500
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Lee; Dongsoo et al.

Integrated laundry processing apparatus and control method thereof

Abstract

A laundry treatment apparatus includes a washing machine module and a drying machine module that are integrally installed in a common cabinet, and a single drying module configured to produce dry hot air. The laundry treatment apparatus has a reduced product size and performs a method for drying laundry in a washing drum and a drying drum. The method includes detecting a dryness in real time while performing drying processes simultaneously in the washing drum and the drying drum by the common drying module, and controlling an amount of supplied dry hot air and an amount of recovered dry hot air to be different to reduce an overall drying process time and electric power consumption. The drying processes performed in the drying machine of the washing machine are simultaneously completed to improve convenience for a user.

Inventors:	Lee; Dongsoo (Seoul, KR), Kim; Taewoong (Seoul, KR), Kang; Hyemin (Seoul, KR)
Applicant:	LG Electronics Inc. (Seoul, KR)
Family ID:	1000008779471
Assignee:	LG Electronics Inc. (Seoul, KR)
Appl. No.:	17/769969
Filed (or PCT Filed):	October 12, 2020
PCT No.:	PCT/KR2020/013872
PCT Pub. No.:	WO2021/075811
PCT Pub. Date:	April 22, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220372685 A1	Nov. 24, 2022

Foreign Application Priority Data

KR10-2019-0129863Oct. 18, 2019

Publication Classification

Int. Cl.: D06F34/26 (20200101); D06F25/00 (20060101); D06F29/00 (20060101); D06F33/63 (20200101); D06F58/24 (20060101); D06F58/26 (20060101); D06F58/38 (20200101); D06F103/34 (20200101); D06F105/32 (20200101)

U.S. Cl.:

CPCD06F34/26 (20200201); D06F25/00 (20130101); D06F29/005 (20130101); D06F33/63 (20200201); D06F58/24 (20130101); D06F58/26 (20130101); D06F58/38 (20200201); D06F2103/34 (20200201); D06F2105/32 (20200201)

Field of Classification Search

CPC:D06F (33/63); D06F (34/26); D06F (25/00); D06F (29/005); D06F (58/38); D06F (58/63); D06F (58/24); D06F (58/26); D06F (2105/32); D06F (2103/34)

USPC:34/282; 34/595-610

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2737799	12/1955	Knipmeyer	68/25	D06F 13/02
3266167	12/1965	Finnegan	361/178	G05D 22/02
4621438	12/1985	Lanciaux	34/604	D06F 58/06
5651192	12/1996	Horwitz	236/78D	D06F 58/30
6493963	12/2001	England	34/575	D06F 58/38
9534329	12/2016	Contarini	N/A	D06F 29/005
10184208	12/2018	Rizzi	N/A	D06F 58/38
11773531	12/2022	Cook, IV	34/491	D06F 58/46
12188172	12/2024	Ryoo	N/A	D06F 58/48
2022/0372685	12/2021	Lee	N/A	D06F 58/38
2024/0328068	12/2023	Choe	N/A	D06F 58/206
2024/0352638	12/2023	Vargas	N/A	D06F 34/26
2025/0034792	12/2024	Chung	N/A	D06F 58/38

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1500932	12/2003	CN	N/A
102758342	12/2011	CN	N/A

208562910	12/2018	CN	N/A
4047118	12/2021	EP	D06F 25/00
4310237	12/2023	EP	D06F 34/18
H08173694	12/1995	JP	N/A
2008307151	12/2007	JP	N/A
20060038796	12/2005	KR	N/A
20210046300	12/2020	KR	N/A
2413806	12/2010	RU	N/A
WO 2007/111413	12/2006	WO	N/A
WO 2019/134478	12/2018	WO	N/A
WO-2021075811	12/2020	WO	D06F 25/00

OTHER PUBLICATIONS

International Search Report in International Appln. No. PCT/KR2020/013872, dated Jan. 19, 2021, 6 pages (with English translation). cited by applicant

Search Report in Russian Appln. No. 2022110364, mailed on Sep. 27, 2022, 4 pages (with English translation). cited by applicant

Extended European Search Report in European Appln. No. 20877790.4, mailed on Oct. 16, 2023, 8 pages. cited by applicant

Primary Examiner: Gravini; Stephen M

Attorney, Agent or Firm: Fish & Richardson P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a National Stage application under 35 U.S.C. § 371 of International Application No. PCT/KR2020/013872, filed on Oct. 12, 2020, which claims the benefit of Korean Application No. 10-2019-0129863, filed on Oct. 18, 2019. The disclosures of the prior applications are incorporated by reference in their entirety.

TECHNICAL FIELD

(2) The present disclosure relates to an integrated laundry treatment apparatus and a method of controlling the same, and more particularly, to an integrated laundry treatment apparatus, in which a washing machine module configured to perform a washing process and a drying machine module configured to perform a drying process are disposed in a common cabinet, and a method of controlling the same.

BACKGROUND

(3) A washing machine refers to a laundry treatment apparatus configured to wash laundry in a drum disposed in a cabinet and configured to be rotated by a drive device. The washing machines may be classified into a top-loading type washing machine and a front-loading type washing machine depending on a posture of the drum.

(4) For instance, the front-loading washing machine has a laid drum, and the top-loading washing machine has an upright drum. The laundry is loaded into the front-loading washing machine through a front portion of the washing machine. The front-loading washing machine may have a smaller size and a larger washing capacity than the top-loading washing machine. In addition, because the front-loading washing machine has high washing performance and does not cause a tangle of laundry, the front-loading washing machine is widely used.

- (5) In addition, a combo-type drying-washing machine, to which a drying function is added by mounting a drying module in the front-loading washing machine, is also widely used.
- (6) In some cases, because the drying module installed in the combo-type drying-washing machine has a limited drying capacity, a large amount of drying time may be taken to dry the entire laundry washed completely.
- (7) Therefore, in some cases, to shorten the drying time, as another laundry treatment apparatus, a drying machine, which has been developed for the purpose of drying the laundry, particularly, a front-loading drying machine similar to the front-loading washing machine, may be used together with the combo-type washing machine.
- (8) In some cases, where the washing machine and the drying machine may be installed together and disposed side by side in a horizontal direction, the washing machine and the drying machine may occupy a large horizontal area.
- (9) In some cases, the washing machine and the drying machine may be stacked in a vertical direction, where, for example, the drying machine is stacked on the washing machine.
- (10) In some cases, where the washing machine and the drying machine are stacked in the vertical direction, independent functional modules, particularly, the drying modules may be provided in the washing machine and the drying machine, which may increase a height of a product in which the washing machine and the drying machine are stacked. As a result, there is a restriction on a position at which the washing machine and the drying machine are installed together. For example, if the entire position at which the washing machine and the drying machine are stacked is higher than a height of the general home, the washing machine and the drying machine may not be installed in the location.
- (11) In some cases, an integrated laundry treatment apparatus may include a washing machine and a drying machine that are assembled and stacked together in a single cabinet, and the washing machine and the drying machine may share some components for implementing functions of a product, thereby reducing an overall size of the product, particularly, an overall height of the product.
- (12) For example, a drive motor and a drying module may be common components for the washing machine and the drying machine. In some cases, the shared drying module includes a common duct, a condenser, a heater, a fan, and a motor which are disposed in the common duct, and inlet and outlet valves configured to switch an outlet and an inlet of the common duct.
- (13) The common drying module may supply hot air to a washing drum and a drying drum by adjusting an inlet valve and an outlet valve to perform drying processes simultaneously in the washing machine and the drying machine, and the common drying module may recover the hot air.
- (14) In some cases, after the laundry in any one of the washing drum and the drying drum is completely dried by the drying processes simultaneously performed in the washing drum and the drying drum, the hot air may continue to be supplied to the drum in which the laundry is completely dried. In some cases, the laundry may be damaged by the hot air, or unnecessary electric power consumption may be incurred.
- (15) In some cases, the drying processes may be performed in the washing drum and the drying drum without monitoring a dried state of the laundry, where the hot air may be continuously supplied and the drum may operate for a preset drying time, which may cause inefficiency in drying time and electric power consumption.

SUMMARY

(16) The present disclosure describes an integrated laundry treatment apparatus, which includes a washing machine module and a drying machine module that are integrally installed in a common cabinet, and a single drying module that is provided to produce dry hot air such that a size of a product can be reduced, where a dried state of laundry in a washing drum and a drying drum is detected in real time while drying processes are simultaneously performed in the washing drum and the drying drum by the common drying module. The amount of supplied dry hot air and the amount

of recovered dry hot air can be differently controlled such that an overall drying process time can be reduced and electric power consumption can be minimized. The drying processes performed in the drying process and the drying machine of the washing machine can be simultaneously completed such that convenience for a user can be improved. The present disclosure also describes a method for controlling the integrated laundry treatment apparatus.

(17) According to one aspect of the subject matter described in this application, a laundry treatment apparatus includes a cabinet that defines an accommodation space therein, a first laundry treatment module disposed in a first portion of the accommodation space, a second laundry treatment module disposed in a second portion of the accommodation space, and a drying module that is configured to produce hot air and has a first outlet and a second outlet that are configured to supply the hot air to the first laundry treatment module and the second laundry treatment module, respectively, and a first inlet and a second inlet that are configured to receive return air from the first laundry treatment module and the second laundry treatment module, respectively. The drying module is configured to absorb at least a part of moisture in the return air. The laundry treatment apparatus further includes an outlet flow path switching valve configured to (i) selectively open one of the first outlet or the second outlet or (ii) simultaneously open both of the first outlet and the second outlet, an inlet flow path switching valve configured to (i) selectively open one of the first inlet or the second inlet or (ii) simultaneously open both of the first inlet and the second inlet, a first humidity sensor configured to measure a first humidity of the return air received through the first inlet, a second humidity sensor configured to measure a second humidity of the return air received through the second inlet, and a controller electrically connected to the first humidity sensor and the second humidity sensor and configured to produce a control signal for operating at least one of the outlet flow path switching valve or the inlet flow path switching valve. The controller is configured to receive a first sensor signal related to the first humidity and a second sensor signal related to the second humidity, the first humidity and the second humidity being measured while the hot air is supplied to the first laundry treatment module and the second laundry treatment module, determine a difference value between the first humidity and the second humidity based on receiving the first sensor signal and the second sensor signal, and based on the difference value exceeding a predetermined threshold value, operate the outlet flow path switching valve to change a first opening degree of the first outlet or a second opening degree of the second outlet, or both.

(18) Implementations according to this aspect can include one or more of the following features. For example, the laundry treatment apparatus can further include a hot air supply duct that includes a first supply duct that connects the first outlet to the first laundry treatment module, and a second supply duct that connects the second outlet to the second laundry treatment module. The laundry treatment apparatus can further include a hot air recovery duct that includes a first recovery duct that connects the first inlet to the first laundry treatment module, and a second recovery duct that connects the second inlet to the second laundry treatment module. In some examples, the first humidity sensor can be disposed in the first recovery duct adjacent to the first inlet, and the second humidity sensor can be disposed in the second recovery duct adjacent to the second inlet.

(19) In some implementations, the drying module can include a common duct that defines a hot air flow path having the first and second inlets and the first and second outlets, a heater disposed in the common duct and configured to produce the hot air, a blower fan disposed in the common duct and configured to blow the hot air toward the first outlet and the second outlet, and a condenser configured to absorb at least a part of the moisture included in the return air received through the first inlet and the second inlet.

(20) In some implementations, the controller can be configured to, before determining the difference value between the first humidity and the second humidity, compare the first humidity and the second humidity to a predetermined target humidity, and based on both of the first humidity and the second humidity exceeding the predetermined target humidity, calculate the difference value between the first humidity and the second humidity. In some examples, the predetermined

target humidity can be a relative humidity of 60%. In some examples, the predetermined threshold value can be a relative humidity of 10%.

(21) In some implementations, the laundry treatment apparatus can include a first actuator electrically connected to the controller and configured to operate the outlet flow path switching valve based on the control signal of the controller, and a second actuator electrically connected to the controller and configured to operate the inlet flow path switching valve based on the control signal of the controller. In some examples, the controller can be configured to, based on the first humidity being greater than the second humidity, produce a first control signal to increase the first opening degree of the first outlet and decrease the second opening degree of the second outlet according to a first predetermined ratio, and transmit the first control signal to the first actuator. In some examples, the controller can be configured to, based on the second humidity being greater than the first humidity, produce a second control signal to increase the second opening degree of the second outlet and decrease the first opening degree of the first outlet according to a second predetermined ratio, and transmit the second control signal to the first actuator.

(22) In some examples, the controller can be configured to, based on the first humidity being greater than the second humidity, produce a third control signal to increase a third opening degree of the first inlet and decrease a fourth opening degree of the second inlet according to a third predetermined ratio, and transmit the third control signal to the second actuator. In some examples, the controller can be configured to, based on the second humidity being greater than the first humidity, produce a fourth control signal to increase the fourth opening degree of the second inlet and decrease the third opening degree of the first inlet according to a fourth predetermined ratio, and transmit the fourth control signal to the second actuator.

(23) According to another aspect, a method controls a laundry treatment apparatus including a cabinet that defines an accommodation space therein, a first laundry treatment module disposed in a first portion of the accommodation space, a second laundry treatment module disposed in a second portion of the accommodation space, and a drying module configured to produce hot air and to supply the hot air to the first laundry treatment module and the second laundry treatment module through a first outlet and a second outlet, respectively. The drying module is further configured to receive return air from the first laundry treatment module and the second laundry treatment module through a first inlet and a second inlet, respectively, and to absorb at least a part of moisture included in the return air. The method includes supplying the hot air to the first and second laundry treatment modules through the first and second outlets, respectively, absorbing at least a part of the moisture included in the return air received through the first and second inlets from the first and second laundry treatment modules, measuring a first humidity of the return air received through the first inlet and a second humidity of the return air received through the second inlet, determining a difference value between the first humidity and the second humidity, determining whether the difference value exceeds a predetermined threshold value, and based on the difference value exceeding the predetermined threshold value, adjusting a ratio of amounts of the hot air to be supplied to the first laundry treatment module and the second laundry treatment module by changing a first opening degree of the first outlet or a second opening degree of the second outlet, or both.

(24) Implementations according to this aspect can include one or more of the following features. For example, the first humidity can be measured from the return air discharged from the first laundry treatment module by a first humidity sensor disposed adjacent to the first inlet, and the second humidity is measured from the return air discharged from the second laundry treatment module by a second humidity sensor disposed adjacent to the second inlet.

(25) In some implementations, the method can include comparing each of the first humidity and the second humidity to a predetermined target humidity, and determining whether each of the first humidity and the second humidity is less than or equal to the predetermined target humidity. In some examples, the difference value can be calculated based on both of the first humidity and the

second humidity exceeding the predetermined target humidity. In some examples, the predetermined target humidity can be a relative humidity of 60%. In some examples, the predetermined threshold value can be a relative humidity of 10%.

(26) In some implementations, adjusting the ratio of the amounts of the hot air can include, based on the first humidity being greater than the second humidity, increasing the first opening degree of the first outlet and decreasing the second opening degree of the second outlet according to a first predetermined ratio, and based on the second humidity being greater than the first humidity, increasing the second opening degree of the second outlet and decreasing the first opening degree of the first outlet according to a second predetermined ratio.

(27) In some implementations, adjusting the ratio of the amounts of the hot air can include, based on the first humidity being greater than the second humidity, increasing a third opening degree of the first inlet and decreasing a fourth opening degree of the second inlet according to a third predetermined ratio, and based on the second humidity being greater than the first humidity, increasing the fourth opening degree of the second inlet and decreasing the third opening degree of the first inlet according to a fourth predetermined ratio.

(28) In some implementations, the method can further include supplying the hot air through the first and second outlets and absorbing at least a part of the moisture in the return air received through the first and second inlets after adjusting the ratio of the amounts of the hot air by changing at least one of the first opening degree of the first outlet, the second opening degree of the second outlet, the third opening degree of the first inlet, or the fourth opening degree of the second outlet.

(29) In some implementations, the method can further include determining whether a predetermined time has elapsed since a start point of supplying the hot air through the first and second outlets after adjusting the ratio of the amounts of the hot air.

(30) In some implementations, the method can further include, based on determining that the predetermined time has elapsed since the start point, repeating operations that include measuring the first humidity and the second humidity, comparing each of the first humidity and the second humidity to the predetermined target humidity, based on both of the first humidity and the second humidity exceeding the predetermined target humidity, calculating the difference value between the first humidity and the second humidity, and adjusting the ratio of the amounts of the hot air by changing at least one of the first opening degree of the first outlet, the second opening degree of the second outlet, the third opening degree of the first inlet, or the fourth opening degree of the second outlet.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a perspective view of an example of an integrated laundry treatment apparatus.

(2) FIG. 2 is a schematic cross-sectional view of FIG. 1.

(3) FIGS. 3 to 6 are schematic configuration views showing examples of a state in which dry hot air is supplied to a drying machine module and a washing machine module and a state in which a supply of dry hot air is cut off in respective process modes.

(4) FIG. 7 is a functional block diagram showing an example of a controller of the integrated laundry treatment apparatus.

(5) FIGS. 8 to 9 are flow charts showing an example of a method for controlling the integrated laundry treatment apparatus.

DETAILED DESCRIPTION

(6) Hereinafter, exemplary implementations of the present disclosure will be described in detail with reference to the accompanying drawings.

(7) FIG. 1 is a perspective view showing an example of an integrated laundry treatment apparatus

1. In some implementations, as illustrated in FIG. 1, the integrated laundry treatment apparatus **1** includes a common cabinet **10** that defines an external appearance of the integrated laundry treatment apparatus **1** and a common accommodation space therein, a first laundry treatment module **20** disposed at an upper side of the common accommodation space, a second laundry treatment module **30** disposed at a lower side of the common accommodation space, and a control panel **40** configured to define a front side of the common cabinet **10** and disposed between the first laundry treatment module **20** and the second laundry treatment module **30**.

(8) For example, the first and second laundry treatment modules **20** and **30**, which are configured to independently perform the laundry treatment processes, can be installed by being integrally stacked in an upward/downward direction in the common cabinet **10**, and the first and second laundry treatment modules **20** and **30** share at least some functional components. Therefore, it is possible to reduce a size, particularly, a size in a height direction of the laundry treatment apparatus in comparison with the related art in which a washing machine and a drying machine are merely stacked.

(9) As illustrated, since the first and second laundry treatment modules **20** and **30** are disposed in the upward/downward direction, a front-loading integrated laundry treatment apparatus is suitable instead of a top-loading integrated laundry treatment apparatus. The first and second laundry treatment modules **20** and **30** respectively include first and second doors **21** and **31** disposed at the front side of the common cabinet **10**.

(10) A first drum **22** for treating the laundry is disposed inside the first door **21**, and a second drum **32** for treating the laundry is disposed inside the second door **31**.

(11) The first and second laundry treatment modules **20** and **30** can perform the same laundry treatment process or different laundry treatment processes.

(12) More specifically, a combination of an upper washing machine module and a lower washing machine module, a combination of an upper drying machine module and a lower drying machine module, a combination of an upper washing machine module and a lower drying machine module, and a combination of an upper drying machine module and a lower washing machine module can be applied.

(13) However, the present disclosure is not limited thereto. Hereinafter, for convenience, an implementation will be described in which a drying machine module **20** is provided as the first laundry treatment module disposed at the upper side, and a washing machine is provided as the second laundry treatment module disposed at the lower side.

(14) The control panel **40** is installed at the front side of the common cabinet **10** and disposed at a position between the drying machine module **20** and the washing machine module **30** in consideration of convenience and accessibility for a user's operation. As illustrated, the control panel **40** includes a manipulation unit **41** into which the user's operation command is inputted, and a display unit **42** configured to display current states of the drying machine module **20** and the washing machine module **30**, a process performing situation, and the like. The control panel **40** is electrically connected to a controller **70** (control unit) to be described below, converts the user's operation command, which is inputted through the manipulation unit **41**, into an electrical signal, and transmits the electrical signal to the controller **70**. For example, the controller **70** can include a processor or an electric circuit.

(15) FIG. 2 is a schematic cross-sectional view of FIG. 1 and illustrates an example of a state in which dry hot air is supplied simultaneously to the drying machine module **20** and the washing machine module **30** through a common drying module **50**.

(16) As illustrated in FIG. 2, the integrated laundry treatment apparatus **1** includes the common drying module **50** configured to supply the hot air simultaneously or independently to the drying machine module **20** and the washing machine module **30** and absorb and remove at least a part of moisture included in the return air recovered from the drying machine module **20** and the washing machine module **30**.

(17) In some implementations, the common drying module **50** can include a common duct **51** having a common flow path therein. The common duct **51** has a first outlet **51c** and a second outlet **51d** provided to supply dry hot air to the drying machine module **20** and the washing machine module **30**, and a first inlet **51a** and a second inlet **51b** provided to recover hot air from the drying machine module **20** and the washing machine module **30**. As illustrated, a hot air supply flow path and a hot air recovery flow path can be disposed between the drying machine module **20** and the washing machine module **30** in the upward/downward direction and optimized to prevent a thermal loss, but the present disclosure is not limited thereto.

(18) The first outlet **51c** of the common drying module **50** is connected to one end of a first supply duct **61** for guiding the hot air discharged from the common duct **51** to the drying machine module **20**, and the second outlet **51d** of the common drying module **50** is connected to one end of a second supply duct **62** for guiding the hot air discharged from the common duct **51** to the washing machine module **30**. That is, as illustrated, the hot air passing through the common duct **51** of the common drying module **50** is divided into hot air supply ducts including the first supply duct **61** and the second supply duct **62**, and the first supply duct **61** and the second supply duct **62** are respectively connected to the drying machine module **20** and the washing machine module **30**.

(19) The other end of the first supply duct **61** is connected to the drying machine module **20**, more specifically, an inlet port **22a** of the drying drum **22**, and the other end of the second supply duct **62** is connected to the washing machine module **30**, more specifically, an inlet port **33a** of a washing machine tub **33**.

(20) The first inlet **51a** of the common drying module **50** is connected to one end of a first recovery duct **63** for guiding the hot air discharged from the drying machine module **20** to the common duct **51**, and the second inlet **51b** of the common drying module **50** is connected to one end of a second recovery duct **64** for guiding the hot air discharged from the washing machine module **30** to the common duct **51**. That is, as illustrated, a flow path structure is formed such that the hot air recovered through the hot air recovery ducts including the first recovery duct **63** and the second recovery duct **64** is merged into the common duct **51**.

(21) The other end of the first recovery duct **63** is connected to a discharge port **22b** of the drying drum **22**, and the other end of the second recovery duct **64** is connected to a discharge port **33b** of the washing machine tub **33**.

(22) In some examples, the integrated laundry treatment apparatus **1** can include a heater **53** configured to produce the hot air and a blower fan **54**, where the heater **53** and the blower fan **54** can be disposed in the common duct **51**. Further, the integrated laundry treatment apparatus **1** can include a condenser **52** disposed at an upstream side of the heater **53** based on a flow direction of the hot air and configured to remove at least a part of moisture included in the return air received from the drying machine module **20** and the washing machine module **30**.

(23) In some implementations, the integrated laundry treatment apparatus **1** can include a blower motor **55** installed outside the common duct **51** and disposed adjacent to the blower fan **54**. The blower motor **55** can be configured to operate the blower fan **54**. As illustrated, the blower fan **54** can receive driving power from the separate blower motor **55** dedicated to operating the blower fan **54**. Alternatively, the blower fan **54** can receive driving power from a drive motor for the washing drum **32** or a drive motor for the drying drum **22**. Hereinafter, for convenience, as illustrated, an implementation will be described in which the blower fan **54** is operated by the blower motor **55** dedicated to operating the blower fan **54**.

(24) For example, the heater **53** can include an electrothermal heater, a gas heater, a heat-pump-type condenser, or the like as a heat source. The condenser **52** can include a water-cooled dehumidification device, a heat-pump-type evaporator, or the like.

(25) Hereinafter, for convenience, the heater **53** and the condenser **52**, which are implemented as a heat pump module **80** will be described, but the present disclosure is not limited thereto.

(26) In some examples, an outlet flow path switching valve **V2** is installed at an outlet side of the

common duct **51** and selectively opens at least any one of the first outlet **51c** and the second outlet **51d** or simultaneously opens the first outlet **51c** and the second outlet **51d**. An inlet flow path switching valve **V1** is installed at an inlet side of the common duct **51** and selectively opens at least any one of the first inlet **51a** and the second inlet **51b** or simultaneously opens the first inlet **51a** and the second inlet **51b**.

(27) As described below, the integrated laundry treatment apparatus **1** has at least four process modes.

(28) The outlet flow path switching valve **V2** serves to switch an outlet flow path of the common drying module **50** so that the hot air is supplied only to at least any one of the first outlet **51c** and the second outlet **51d** of the common duct **51** or the hot air is supplied to both the first outlet **51c** and the second outlet **51d** for the respective process modes.

(29) Likewise, the inlet flow path switching valve **V1** serves to switch an inlet flow path of the common drying module **50** so that the hot air is recovered only through at least any one of the first inlet **51a** and the second inlet **51b** of the common duct **51** or the hot air is recovered through both the first inlet **51a** and the second inlet **51b** for the respective process modes.

(30) In addition, as described below, the outlet flow path switching valve **V2** and the inlet flow path switching valve **V1** can adjust opening degrees of the first and second inlets **51a** and **51b** and the first and second outlets **51c** and **51d** to control a flow rate of the hot air to be supplied to the drying machine module **20** and the washing machine module **30** in the process mode in which the hot air is supplied to both the drying machine module **20** and the washing machine module **30**.

(31) In some examples, the outlet flow path switching valve **V2** and the inlet flow path switching valve **V1** are respectively connected to valve actuators **A1** and **A2** for operating the outlet flow path switching valve **V2** and the inlet flow path switching valve **V1**, and valve positions are controlled by the valve actuators **A1** and **A2**, respectively.

(32) For example, the outlet flow path switching valve **V2** and the inlet flow path switching valve **V1** are each provided in the form of a three-way valve. However, a various types of devices, which are capable of switching a 3-way flow path and adjusting opening degrees of the inlet and the outlet of the common drying module **50** for the respective process modes, can be applied as the outlet flow path switching valve **V2**, the inlet flow path switching valve **V1**, and the valve actuators **A1** and **A2**. The modified examples belong to the scope of the present disclosure.

(33) The detailed description of valve position control for the respective process modes will be made with reference to FIGS. **3** to **6**.

(34) Further, the integrated laundry treatment apparatus **1** includes a first humidity sensor **SH1** configured to measure a first humidity of the hot air recovered through the first inlet **51a**, and a second humidity sensor **SH2** configured to measure a second humidity of the hot air recovered through the second inlet **51b**.

(35) The first humidity sensor **SH1** serves to check a drying process state of the laundry in the drying drum **22** by measuring the humidity of the hot air discharged from the drying drum **22** of the drying machine module **20**. The second humidity sensor **SH1** serves to check a drying process state of the laundry in the washing drum **32** by measuring the humidity of the hot air discharged from the washing drum **32** and the tub **33** of the washing machine module **30**.

(36) The first humidity sensor **SH1** and the second humidity sensor **SH1** are electrically connected to the controller **70**. A signal related to the first humidity measured by the first humidity sensor **SH1** and a signal related to the second humidity measured by the second humidity sensor **SH1** are periodically transmitted to the controller **70**.

(37) Therefore, the controller **70** can check the dried state of the laundry in the drying drum **22** and the washing drum **32** based on the first humidity and the second humidity while the drying machine module **20** and the washing machine module **30** simultaneously perform the drying processes. The controller **70** creates a control signal for switching the flow path based on the received first and second humidities so that a large amount of hot air can be supplied to a portion where the laundry

is less dried, i.e., a portion where the hot air having a higher humidity is discharged. The controller **70** transmits the control signal to the valve actuators **A1** and **A2**, and the actuators **A1** and **A2** operate the valves based on the corresponding control signal and adjust the opening degrees of the first and second inlets **51a** and **51b** and the opening degrees of the first and second outlets **51c** and **51d**.

(38) The detailed configuration related to the process of controlling the opening degrees of the first and second inlets **51a** and **51b** and the opening degrees of the first and second outlets **51c** and **51d** based on the first and second humidities while the drying processes are simultaneously performed will be described below with reference to FIGS. **8** and **9**.

(39) In some examples, since the first humidity sensor **SH1** and the second humidity sensor **SH1** serve to check the dried state of the laundry in the drying drum **22** and the dried state of the laundry in the washing drum **32**, the first humidity sensor **SH1** and the second humidity sensor **SH1** need to measure a state before a recovery flow path from the drying machine module **20** and a recovery flow path from the washing machine module **30** are merged. Therefore, the first humidity sensor **SH1** and the second humidity sensor **SH1** can be respectively installed in the first recovery duct **63** and the second recovery duct **64** and disposed adjacent to the first inlet **51a** and the second inlet **51b** at the upstream positions from the inlet flow path switching valve **V1** based on the flow direction of the hot air.

(40) The first humidity sensor **SH1** and the second humidity sensor **SH1** can be configured to measure a relative humidity in the air and convert the measured humidity into an electrical signal.

(41) FIGS. **3** to **6** are schematic configuration views for explaining a state in which the dry hot air is supplied to the drying machine module **20** and the washing machine module **30** or a state in which the supply of the dry hot air is cut off for respective process modes.

(42) As described above, the integrated laundry treatment apparatus **1** can have at least four process modes including first to fourth process modes based on supplied and recovered states of the dry hot air.

(43) The first process mode is a process mode in which the drying machine module **20** performs the drying process singly and the washing machine module **30** is in a stopped state or performs the washing process. In the first process mode, the dry hot air is supplied only to the drying machine module **20**.

(44) The second process mode is a process mode in which the washing machine module **30** performs the drying process singly and the drying machine module **20** is in a stopped state. In the second process mode, the dry hot air is supplied only to the washing machine module **30**.

(45) The third process mode is a process mode in which the drying machine module **20** performs the drying process and, at the same time, the washing machine module **30** performs the washing process and then performs the drying process. In the third process mode, the dry hot air is supplied only to the drying machine module **20** while the washing machine module **30** performs the washing process. When the washing machine module **30** completes the washing process and performs the drying process, the hot air supply and recovery flow paths are switched by the outlet flow path switching valve **V2** and the inlet flow path switching valve **V1** so that the hot air can be simultaneously supplied to the drying machine module **20** and the washing machine module **30**.

(46) The fourth process mode is a process mode in which the drying machine module **20** and the washing machine module **30** simultaneously perform the drying processes and the dry hot air is simultaneously supplied to the drying machine module **20** and the washing machine module **30**.

(47) FIG. **3** illustrates an example of a state in which the dry hot air is supplied only to the drying machine module **20** in the first and third process modes. Referring to FIG. **3**, the washing machine module **30** is in the stopped state in the first process mode, and the washing machine module **30** performs the washing process in the third process mode. Therefore, the dry hot air is not supplied to the washing machine module **30**, the total amount of dry hot air is supplied only to the drying machine module **20**, and the dry hot air is recovered from the drying machine module **20**.

(48) The outlet flow path switching valve V2 is operated by the valve actuator A2 to fully open the first outlet 51c of the common duct 51 and fully close the second outlet 51d of the common duct 51, and the inlet flow path valve is operated by the valve actuator A1 to fully open the first inlet 51a of the common duct 51 and fully close the second inlet 51b of the common duct 51, such that the dry hot air can be supplied only to the drying machine module 20 and the drying machine module 20 can perform the drying process singly. Therefore, as illustrated in FIG. 3, a single hot air circulation flow path is formed in respect to the drying machine module 20.

(49) FIG. 3 illustrates an implementation in which the outlet flow path switching valve V2 and the inlet flow path switching valve V1, which are each a three-way valve, are each provided in the form of a flap valve. Hereinafter, for convenience, the implementation will be described in which the outlet flow path switching valve V2 and the inlet flow path switching valve V1 are each provided in the form of a flap valve, but the present disclosure is not limited thereto.

(50) In some examples, the dried state of the laundry in the drying drum 22 is monitored in real time by the first humidity sensor SH1 while the dry hot air is supplied only to the drying machine module 20 by the common drying module 50 in the first and third process modes.

(51) Therefore, as described below, when the controller 70 determines that the first humidity measured by the first humidity sensor SH1 reaches a relative humidity of 60%, for example, the controller 70 can perform control to complete the drying process in the drying machine module 20, thereby preventing the laundry from being excessively dried and reducing electric power consumption.

(52) FIG. 4 illustrates an example of a state in which the dry hot air is supplied only to the washing machine module 30 in the second process mode. Referring to FIG. 4, since the drying machine module 20 is in the stopped state in the second process mode, the dry hot air is not supplied to the drying machine module 20, and the total amount of dry hot air is supplied only to the washing machine module 30.

(53) The outlet flow path switching valve V2 is operated by the valve actuator A2 to fully close the first outlet 51c of the common duct 51 and fully open the second outlet 51d of the common duct 51, and the inlet flow path valve is operated by the valve actuator A1 to fully close the first inlet 51a of the common duct 51 and fully open the second inlet 51b of the common duct 51, such that the dry hot air can be supplied only to the washing machine module 30 and the washing machine module 30 can perform the drying process singly. Therefore, as illustrated in FIG. 4, a single hot air circulation flow path is formed in respect to the washing machine module 30.

(54) The dried state of the laundry in the washing drum 32 is monitored in real time by the second humidity sensor SH1 in the same way as the first and third process modes while the dry hot air is supplied only to the washing machine module 30 by the common drying module 50 in the second process mode.

(55) Therefore, as described below, when the controller 70 determines that the second humidity measured by the second humidity sensor SH1 reaches a relative humidity of 60%, for example, the controller 70 performs control to complete the drying process in the washing machine module 30, thereby preventing the laundry from being excessively dried and reducing electric power consumption.

(56) FIG. 5 illustrates an example of a state in which the dry hot air is simultaneously supplied to the drying machine module 20 and the washing machine module 30 in the third and fourth process modes.

(57) Referring to FIG. 5, the dry hot air is simultaneously supplied to the drying machine module 20 and the washing machine module 30 and simultaneously recovered from the drying machine module 20 and the washing machine module 30 so that the washing process of the washing machine module 30 is completed and the drying process is performed while the drying machine module 20 performs the drying process in the third process mode or so that both the drying machine module 20 and the washing machine module 30 perform the drying processes in the fourth

process mode.

(58) The outlet flow path switching valve V2 is operated by the valve actuator A2 to at least partially open the first outlet 51c and the second outlet 51d of the common duct 51, and the inlet flow path valve is operated by the valve actuator A1 to at least partially open the first inlet 51a and the second inlet 51b of the common duct 51, such that the dry hot air can be simultaneously supplied to the drying machine module 20 and the washing machine module 30, and the drying machine module 20 and the washing machine module 30 can simultaneously perform the drying processes. Therefore, as illustrated in FIG. 4, hot air circulation flow paths are formed in respect to the drying machine module 20 and the washing machine module 30.

(59) As illustrated in FIG. 5, the outlet flow path switching valve V2 is controlled so that the first outlet 51c and the second outlet 51d of the common duct 51 have the same opening degree, and the inlet flow path switching valve V1 is controlled so that the first inlet 51a and the second inlet 51b of the common duct 51 have the same opening degree, such that the dry hot air is supplied basically at the same flow rate to the drying drum 22 and the washing drum 32 while the drying drum 22 and the washing drum 32 simultaneously perform the drying process.

(60) However, in a situation in which a large amount of dry hot air needs to be supplied to the drying machine module 20 such as a case in which the amount of laundry in the drying drum 22 is larger than the amount of laundry in the washing drum 32 and a case in which the washing drum 32 is sterilized and disinfected, the outlet flow path switching valve V2 and the inlet flow path switching valve V1 can be controlled so that the opening degree of the first inlet 51a is larger than the opening degree of the second inlet 51b and the opening degree of the first outlet 51c is larger than the opening degree of the second outlet 51d, such that a larger amount of dry hot air can be supplied to the drying drum 22 and recovered from the drying drum 22.

(61) In some examples, the dried state of the laundry in the drying drum 22 and the dried state of the laundry in the washing drum 32 are respectively monitored in real time by the first humidity sensor SH1 and the second humidity sensor SH1 while the dry hot air is simultaneously supplied to the drying machine module 20 and the washing machine module 30 by the common drying module 50 while the drying machine module 20 and the washing machine module 30 simultaneously perform the drying processes.

(62) As described below, when the controller 70 determines that at least any one of the first humidity and the second humidity measured by the first humidity sensor SH1 and the second humidity sensor SH1 reaches a predetermined relative humidity, for example, when the controller 70 determines that the first humidity reaches a relative humidity of 60% and the second humidity is lower than a relative humidity of 60%, the controller 70 can switch the outlet flow path switching valve V2 and the inlet flow path switching valve V1 to the states illustrated in FIG. 4 so that the drying process in the drying machine module 20 is stopped and the total amount of dry hot air is transmitted to the washing machine module 30, thereby preventing the laundry in the drying drum 22 from being excessively dried and reducing electric power consumption.

(63) In some examples, when the controller 70 determines that the second humidity reaches a relative humidity of 60% and the first humidity is lower than a relative humidity of 60%, the controller 70 switches the outlet flow path switching valve V2 and the inlet flow path switching valve V1 to the states illustrated in FIG. 3 so that the drying process in the washing machine module 30 is stopped and the total amount of dry hot air is transmitted to the drying machine module 20.

(64) In some examples, there can occur a case in which the simultaneous drying processes are entirely further delayed than expected when a deviation between the first humidity and the second humidity is severe during the simultaneous drying processes even though both the first humidity and the second humidity are lower than the relative humidity of 60%.

(65) To prevent the delay, the controller 70 calculates a difference value between the first humidity and the second humidity. When the controller 70 determines that the corresponding difference

value exceeds a predetermined threshold value, particularly, a relative humidity of 10%, the controller **70** performs control to switch the state of the outlet flow path switching valve **V2** and the state of the inlet flow path switching valve **V1** so that a larger amount of hot air can be supplied to one of the drying machine module **20** and the washing machine module **30** from which the hot air having a higher humidity is discharged.

(66) For example, when the controller **70** determines that the first humidity is higher than the second humidity by a relative humidity of 10%, the controller **70** controls the outlet flow path switching valve **V2** to increase the opening degree of the first outlet **51c** of the common duct **51** by a predetermined ratio and decrease the opening degree of the second outlet **51d** by the predetermined ratio, and the controller **70** controls the inlet flow path switching valve **V1** to adjust the opening degrees of the first inlet **51a** and the second inlet **51b** of the common duct **51** by the same predetermined ratio.

(67) In this case, the predetermined ratio by which the opening degrees of the first and second inlets **51a** and **51b** and the first and second outlets **51c** and **51d** of the common duct **51** are adjusted can be about 5%.

(68) As described above, the opening degrees of the first and second inlets **51a** and **51b** and the first and second outlets **51c** and **51d** of common duct **51** can be adjusted and the amount of dry hot air can be differently controlled based on the deviation between the first humidity and the second humidity, thereby minimizing electric power consumption caused when the simultaneous drying processes are entirely delayed.

(69) The detailed contents related to the measurement of the first humidity and the second humidity during the simultaneous drying processes and the detailed contents related to control of the outlet flow path switching valve **V2** and the inlet flow path switching valve **V1** will be described below with reference to FIG. **9A** and FIG. **9B**.

(70) FIG. **6** illustrates an example of the integrated laundry treatment apparatus **1** in which the inlet flow path switching valve **V1** and the outlet flow path switching valve **V2** each include two valves.

(71) As illustrated in FIG. **6**, the inlet flow path switching valve **V1** includes a first inlet flow path switching valve **V11** dedicated to opening or closing the first inlet **51a**, and a second inlet flow path switching valve **V12** dedicated to opening or closing the second inlet **51b**.

(72) Further, the outlet flow path switching valve **V2** includes a first outlet flow path switching valve **V21** dedicated to opening or closing the first outlet **51c**, and a second outlet flow path switching valve **V22** dedicated to opening or closing the second outlet **51d**.

(73) In some examples, as shown in FIGS. **3** to **5**, the first and second inlets **51a** and **51b** may not be simultaneously closed and the first, and second outlets **51c** and **51d** may not be simultaneously closed. In some examples, as shown in FIG. **6**, the drying machine hot air flow paths and the washing machine hot air flow paths can be simultaneously closed.

(74) Since the hot air flow paths are simultaneously closed as described above, it is possible to remove lint and dust produced during the drying processes in the drying machine module **20** and the washing machine module **30** and accumulated on the condenser **52**. Further, it is possible to prevent lint and dust, which remain while the condenser **52** is cleaned, from entering the washing drum **32** or the drying drum **22** again and contaminating the washing drum **32** or the drying drum **22**.

(75) The first outlet flow path switching valve **V21** and the second outlet flow path switching valve **V22** are respectively operated by dedicated valve actuators **V21** and **V22**, and the first inlet flow path switching valve **V11** and the second inlet flow path switching valve **V12** are respectively operated by dedicated valve actuators **V11** and **V12**.

(76) FIG. **7** is an example of a functional block diagram representing the controller **70** of the integrated laundry treatment apparatus **1**, and FIGS. **8** to **9** are flow charts related to an example method for controlling the integrated laundry treatment apparatus **1**.

(77) Hereinafter, the method of controlling the integrated laundry treatment apparatus **1** will be

described focusing on the controller **70** illustrated in FIG. 7.

(78) In some implementations, as illustrated in FIG. 8, the controller can perform a process mode selection signal receiving step **S1** of receiving a signal for selecting at least any one of the process modes at least including the first to fourth process modes.

(79) In the process mode selection signal receiving step **S1**, the controller **70** receives the signal for selecting the first to fourth process modes based on a selection command physically inputted from the manipulation unit **41** of the control panel **40** illustrated in FIG. 7. Alternatively, the controller **70** can receive the signal for selecting any one of the first to fourth process modes based on a selection command inputted in a wireless manner through a communication unit **CM** instead of the control panel **40**. The communication unit **CM** is connected to a user mobile device through any wireless communication.

(80) Then, the controller **70** performs a process mode determining step **S2** of determining which process mode is selected among the first to fourth process modes based on the received selection signal.

(81) In this case, the controller **70** can provide the user with information indicating that a particular process mode is selected through the user mobile device or the display unit **42** illustrated in FIG. 7.

(82) When the determination result indicates that the first process mode is selected in the process mode determining step **S2**, the controller **70** performs a drying machine hot air flow path opening step **S3** of opening the drying machine hot air flow path as a preparation step for allowing the drying machine module **20** to perform the drying process singly.

(83) In the drying machine hot air flow path opening step **S3**, the controller **70** produces the control signal for fully opening the first inlet **51a** and the first outlet **51c** of the common duct **51** and fully closing the second inlet **51b** and the second outlet **51d** of the common duct **51**, like the state illustrated in FIG. 3, and transmits the control signal to the valve actuators **A1** and **A2** illustrated in FIG. 7.

(84) When the drying machine hot air flow path opening step **S3** is completed, the controller **70** performs a drying machine drying process performing step **S4** of performing the drying process in the drying machine module **20**.

(85) In the drying machine drying process performing step **S4**, the controller **70** can check whether the first door **21** of the drying machine module **20** is closed from a sensor for the first door **21** of a sensor unit **SM** illustrated in FIG. 7 before supplying electric power to a first drive unit **91** of a drive unit **90** for operating the drying drum **22**.

(86) When the checking result indicates that the first door **21** is closed, the controller **70** can detect the amount of laundry by initially operating the drying drum **22** by supplying electric power from a power source unit **PM** to the first drive unit **91**.

(87) The controller **70** can compare the detected amount of laundry with drying time data for the respective amounts of laundry stored in advance in a memory **71** and transmit information on expected drying process time to the display unit **42** or the user mobile device.

(88) In this case, when the expected drying process time is determined by initially operating the first drive unit **91**, the controller **70** performs control to produce the dry hot air by supplying the electric power to the heat pump module **80** and the blower motor **55** of the common drying module **50**.

(89) In some examples, since the washing machine module **30** is in a stopped state in the first process mode, the controller **70** performs control not to supply electric power from the power source unit **PM** to a second drive unit **92** of the drive unit **90** for operating the washing drum **32**.

(90) In some examples, when the determination result indicates that the second process mode is selected in the process mode determining step **S2**, the controller **70** performs a washing machine hot air flow path opening step **S5** of opening the washing machine hot air flow path as a preparation step of allowing the washing machine module **30** to perform the drying process singly, as illustrated in FIG. 8.

- (91) In the washing machine hot air flow path opening step S5, the controller **70** produces the control signal for fully opening the second inlet **51b** and the second outlet **51d** of the common duct **51** and fully closing the first inlet **51a** and the first outlet **51c** of the common duct **51**, like the state illustrated in FIG. 4, and transmits the control signal to the valve actuators **A1** and **A2** illustrated in FIG. 7.
- (92) When the washing machine hot air flow path opening step S5 is completed, the controller **70** performs a washing machine drying process performing step S6 of performing the drying process in the washing machine module **30**.
- (93) In the washing machine drying process performing step S6, the controller **70** can check whether the second door **31** of the washing machine module **30** is closed from a sensor for the second door **31** illustrated in FIG. 7 before supplying electric power to the second drive unit **92** for operating the washing drum **32**.
- (94) When the checking result indicates that the second door **31** is closed, the controller **70** can detect the amount of laundry by initially operating the washing drum **32** by supplying electric power from the power source unit PM to the second drive unit **92**.
- (95) The controller **70** can compare the detected amount of laundry with drying time data for the respective amounts of laundry stored in advance in a memory **71** and transmit information on expected drying process time to the display unit **42** or the user mobile device. The expected drying process time of the washing machine module **30** can be different from the drying process time of the drying machine module **20**.
- (96) In this case, when the expected drying process time is determined by initially operating the second drive unit **92**, the controller **70** performs control to produce the dry hot air by supplying the electric power to the heat pump module **80** and the blower motor **55** of the common drying module **50**.
- (97) In some examples, since the drying machine module **20** is in a stopped state in the second process mode, the controller **70** performs control not to supply electric power from the power source unit PM to the first drive unit **91** for operating the drying drum **22**.
- (98) FIG. 9A and FIG. 9B illustrate example steps to be performed by performing the third or fourth process mode among the first to fourth process modes.
- (99) As illustrated in FIG. 9A and FIG. 9B, when the determination result indicates that the third process mode is selected in the process mode determining step S2, the controller **70** performs a drying machine hot air flow path opening step S7 of opening the drying machine hot air flow path as a preparation step for allowing the drying machine module **20** to perform the drying process singly.
- (100) The drying machine hot air flow path opening step S7 is performed basically in the same way as the drying machine hot air flow path opening step S3 performed when the first process mode is selected. The controller **70** produces the control signal for fully opening the first inlet **51a** and the first outlet **51c** of the common duct **51** and fully closing the second inlet **51b** and the second outlet **51d** of the common duct **51** so that the hot air flow path is created only for the drying machine module **20**, and the controller **70** can transmit the control signal to the valve actuators **A1** and **A2** illustrated in FIG. 7.
- (101) When the drying machine hot air flow path opening step S7 is completed, the controller **70** can perform a drying machine drying/washing machine washing process performing step S8 of performing the drying process in the drying machine module **20** and the washing process in the washing machine module **30**.
- (102) In the drying machine drying/washing machine washing process performing step S8, the controller **70** can check whether the first door **21** and the second door **31** are closed from the sensor for the first door **21** and the sensor for the second door **31** illustrated in FIG. 7 before supplying electric power to the first drive unit **91** and the second drive unit **92**.
- (103) When the checking result indicates that the first door **21** and the second door **31** are closed,

the controller **70** can detect the amount of laundry in the drying drum **22** and the amount of laundry in the washing drum **32** by initially operating the drying drum **22** and the washing drum **32** by supplying electric power from the power source unit PM to the first drive unit **91** and the second drive unit **92**.

(104) The controller **70** can compare the detected amount of dried laundry with drying time data for the respective amounts of dried laundry stored in advance in the memory **71** and transmits information on expected drying process time of the drying machine module **20** to the display unit **42** or the user mobile device. Further, the controller **70** can compare the detected amount of washed laundry with drying time data for the respective amounts of washed laundry stored in advance in the memory **71** and transmit information on expected washing process time of the washing machine module **30** to the display unit **42** or the user mobile device.

(105) Thereafter, when the expected drying process time of the drying machine module **20** is determined by initially operating the first drive unit **91**, the controller **70** performs control to produce the dry hot air by supplying the electric power to the heat pump module **80** and the blower motor **55** of the common drying module **50**.

(106) Next, the controller **70** performs a simultaneous drying processes entering determining step **S9** of determining whether the washing process in the washing machine module **30** is completed, whether to supply the dry hot air to the washing machine module **30**, and whether the simultaneous drying processes enter.

(107) More specifically, whether the washing process in the washing machine module **30** is completed is determined during the drying process in the drying machine module **20**.

(108) If the controller **70** determines that the drying process in the drying machine module **20** is completed in a state in which the washing process in the washing machine module **30** is not completed, the controller **70** stops the drying process in the drying machine module **20** and continuously performs the washing process in the washing machine module **30**.

(109) In addition, when the washing process in the washing machine module **30** is completed and the drying process in the washing machine module **30** is intended to be performed after the drying process in the drying machine module **20** is completed, the controller **70** performs control to perform the washing machine hot air flow path opening step **S5** of opening the washing machine hot air flow path illustrated in FIG. **8** and perform the washing machine drying process performing step **S6**.

(110) In some examples, when the washing process in the washing machine module **30** is completed and the drying process in the washing machine module **30** is intended to be performed in a state in which the drying process in the drying machine module **20** is being performed, the controller **70** performs a drying machine/washing machine hot air flow path simultaneous opening step **S10** of simultaneously opening the drying machine hot air flow path and the washing machine hot air flow path as a preparation step for simultaneously performing the drying processes in the drying machine module **20** and the washing machine module **30**.

(111) In the drying machine/washing machine hot air flow path simultaneous opening step **S10**, the controller **70** can produce the control signal for fully opening the first and second inlets **51a** and **51b** and the first and second outlets **51c** and **51d** of the common duct **51** so that the hot air flow paths can be simultaneously created for the drying machine module **20** and the washing machine module **30**, and the controller **70** can transmit the control signal to the valve actuators **A1** and **A2** illustrated in FIG. **7**.

(112) As illustrated in FIG. **5**, the outlet flow path switching valve **V2** is operated by the valve actuator **A2** to at least partially open the first outlet **51c** and the second outlet **51d** of the common duct **51**, and the inlet flow path valve is operated by the valve actuator **A1** to at least partially open the first inlet **51a** and the second inlet **51b** of the common duct **51**, such that the dry hot air can be simultaneously supplied to the drying machine module **20** and the washing machine module **30**, and the drying machine module **20** and the washing machine module **30** can simultaneously

perform the drying processes.

(113) As described above, the outlet flow path switching valve V2 is controlled so that the first outlet 51c and the second outlet 51d of the common duct 51 have the same opening degree, and the inlet flow path switching valve V1 is controlled so that the first inlet 51a and the second inlet 51b of the common duct 51 have the same opening degree, such that the dry hot air is supplied basically at the same flow rate to the drying drum 22 and the washing drum 32 while the drying drum 22 and the washing drum 32 simultaneously perform the drying process.

(114) In some examples, when the drying machine/washing machine hot air flow path simultaneous opening step S10 is completed as described above, the controller 70 performs a drying machine/washing machine simultaneous drying processes performing step S11 of performing the drying processes simultaneously in the drying machine module 20 and the washing machine module 30.

(115) During the simultaneous drying processes performing step S11, the controller 70 performs a humidity measuring step S12 of measuring, in real time, a relative humidity of the hot air recovered through the first inlet 51a and the second inlet 51b of the common drying module 50.

(116) More specifically, in the humidity measuring step S12, the controller 70 performs a first humidity measuring step of measuring, in real time, the first humidity by using the first humidity sensor SH1 to check the dried state of the laundry in the drying drum 22, and a first humidity measuring step of measuring, in real time, the second humidity by using the second humidity sensor SH1 to check the dried state of the laundry in the washing drum 32.

(117) In this case, when the controller 70 determines that at least any one of the first humidity and the second humidity measured by the first humidity sensor SH1 and the second humidity sensor SH1 reaches a predetermined target humidity, for example, when the controller 70 determines that the first humidity reaches a relative humidity of 60% and the second humidity is lower than a relative humidity of 60%, the controller 70 can switch the outlet flow path switching valve V2 and the inlet flow path switching valve V1 to the states illustrated in FIG. 4 so that the drying process in the drying machine module 20 is stopped and the total amount of dry hot air is transmitted to the washing machine module 30, thereby preventing the laundry in the drying drum 22 from being excessively dried and reducing electric power consumption.

(118) In some examples, when the controller 70 determines that the second humidity reaches a relative humidity of 60% and the first humidity is lower than a relative humidity of 60%, the controller 70 performs control to switch the outlet flow path switching valve V2 and the inlet flow path switching valve V1 to the states illustrated in FIG. 3 so that the drying process in the washing machine module 30 is stopped and the total amount of dry hot air is transmitted to the drying machine module 20.

(119) In some examples, there can occur a case in which the simultaneous drying processes are entirely further delayed than expected when a deviation between the first humidity and the second humidity is severe during the simultaneous drying processes even though both the first humidity and the second humidity are lower than the relative humidity of 60%.

(120) In some implementations, the controller 70 can perform a humidity difference value determining step S13 of calculating a difference value between the first humidity and the second humidity and determining whether the corresponding difference value exceeds a predetermined threshold value, particularly, a relative humidity of 10%.

(121) In the humidity difference value determining step S13, when the determination result indicates that the difference value between the first humidity and the second humidity does not exceed the predetermined threshold value, the controller 70 perform control to continuously perform the drying time drying machine/washing machine simultaneous drying processes performing step S11 until the first humidity and the second humidity reach a target humidity.

(122) However, in the humidity difference value determining step S13, when the determination result indicates that the difference value between the first humidity and the second humidity

exceeds the predetermined threshold value, the controller 70 performs a flow path opening ratio adjusting step S14 of adjusting a ratio of the hot air supplied to the drying machine module 20 and the washing machine module 30 by changing the opening degrees of the first outlet 51c and the second outlet 51d of the common duct 51.

(123) In the flow path opening ratio adjusting step S14, the controller 70 performs control to switch the state of the outlet flow path switching valve V2 and the state of the inlet flow path switching valve V1 so that a larger amount of hot air can be supplied to one of the drying machine module 20 and the washing machine module 30 from which the hot air having a higher humidity is discharged.

(124) For example, when the controller 70 determines that the first humidity is higher than the second humidity by a relative humidity of 10%, the controller 70 controls the outlet flow path switching valve V2 to increase the opening degree of the first outlet 51c of the common duct 51 by a predetermined ratio and decrease the opening degree of the second outlet 51d by the predetermined ratio, and the controller 70 controls the inlet flow path switching valve V1 to adjust the opening degrees of the first inlet 51a and the second inlet 51b of the common duct 51 by the same predetermined ratio.

(125) In this case, the predetermined ratio by which the opening degrees of the first and second inlets 51a and 51b and the first and second outlets 51c and 51d of the common duct 51 are adjusted can be about 5%.

(126) In some examples, in a case in which the difference value between the first humidity and the second humidity is a difference value larger than the predetermined threshold value, i.e., a relative humidity of 10%, for example, a difference value equal to or larger than a relative humidity of 15%, a drum rotational speed adjusting step S15 of increasing a rotational speed of the drum of one of the drying machine module 20 and the washing machine module 30 from which the hot air having a higher humidity is discharged can be performed in addition to the flow path opening ratio adjusting step S14.

(127) The controller 70 continuously performs the drying machine/washing machine simultaneous drying processes performing step S16 in the state in which the opening ratio of the flow path and the rotational speed of the drum are adjusted as described above, and the controller 70 performs a time elapse determining step 17 of determining whether a predetermined time elapses after the adjustment of the opening ratio of the flow path and the rotational speed of the drum is completed. The predetermined time can be set as 5 minutes to 10 minutes in consideration of the amount of laundry existing in the drying drum 22 and the washing drum 32 and the current remaining drying process time.

(128) When the determination result indicates that the predetermined time has elapsed in the time elapse determining step 17, the controller 70 performs control to perform the humidity measuring step S12 again and end the drying process or perform the humidity difference value determining step S13, the flow path opening ratio adjusting step S14, the drum rotational speed adjusting step S15, and the drying machine/washing machine simultaneous drying processes performing step S16 again depending on the humidity measurement result.

(129) In some examples, as illustrated in FIG. 9A and FIG. 9B, when the determination result indicates that the fourth process mode is selected in the process mode determining step S2, the remaining steps, except for the drying machine hot air flow path opening step S7, the drying machine drying/washing machine washing process performing step S8, the simultaneous drying processes entering determining step S9 performed in the third process mode, are performed. Therefore, the controller 70 performs control to immediately perform the drying machine/washing machine hot air flow path simultaneous opening step S10 simultaneously on the drying machine module 20 and the washing machine module 30 and then perform the above-mentioned subsequent steps.

(130) It can be understood that the above-mentioned technical features of the present disclosure can be carried out in any other specific form by those skilled in the art without changing the technical

spirit or the essential features of the present disclosure.

(131) Accordingly, it should be understood that the aforementioned implementations are described for illustration in all aspects and are not limited, and the scope of the present disclosure shall be represented by the claims to be described below, and it should be construed that all of the changes or modified forms derived from the meaning and the scope of the claims, and an equivalent concept thereto are included in the scope of the present disclosure.

Claims

1. A laundry treatment apparatus comprising: a cabinet that defines an accommodation space therein; a first laundry treatment module disposed in a first portion of the accommodation space; a second laundry treatment module disposed in a second portion of the accommodation space; a drying module configured to produce hot air, the drying module having: a first outlet and a second outlet that are configured to supply the hot air to the first laundry treatment module and the second laundry treatment module, respectively, and a first inlet and a second inlet that are configured to receive return air from the first laundry treatment module and the second laundry treatment module, respectively, the drying module being configured to absorb at least a part of moisture in the return air; an outlet flow path switching valve configured to (i) selectively open one of the first outlet or the second outlet or (ii) simultaneously open both of the first outlet and the second outlet; an inlet flow path switching valve configured to (i) selectively open one of the first inlet or the second inlet or (ii) simultaneously open both of the first inlet and the second inlet; a first humidity sensor configured to measure a first humidity of the return air received through the first inlet; a second humidity sensor configured to measure a second humidity of the return air received through the second inlet; and a controller electrically connected to the first humidity sensor and the second humidity sensor and configured to produce a control signal for operating at least one of the outlet flow path switching valve or the inlet flow path switching valve, wherein the controller is configured to: receive a first sensor signal related to the first humidity and a second sensor signal related to the second humidity, the first humidity and the second humidity being measured while the hot air is supplied to the first laundry treatment module and the second laundry treatment module, determine a difference value between the first humidity and the second humidity based on receiving the first sensor signal and the second sensor signal, and based on the difference value exceeding a predetermined threshold value, operate the outlet flow path switching valve to change a first opening degree of the first outlet or a second opening degree of the second outlet, or both.
2. The laundry treatment apparatus of claim 1, further comprising: a hot air supply duct comprising: a first supply duct that connects the first outlet to the first laundry treatment module, and a second supply duct that connects the second outlet to the second laundry treatment module; and a hot air recovery duct comprising: a first recovery duct that connects the first inlet to the first laundry treatment module, and a second recovery duct that connects the second inlet to the second laundry treatment module.
3. The laundry treatment apparatus of claim 2, wherein the first humidity sensor is disposed in the first recovery duct adjacent to the first inlet, and wherein the second humidity sensor is disposed in the second recovery duct adjacent to the second inlet.
4. The laundry treatment apparatus of claim 1, wherein the drying module comprises: a common duct that defines a hot air flow path having the first and second inlets and the first and second outlets; a heater disposed in the common duct and configured to produce the hot air; a blower fan disposed in the common duct and configured to blow the hot air toward the first outlet and the second outlet; and a condenser configured to absorb at least a part of the moisture included in the return air received through the first inlet and the second inlet.
5. The laundry treatment apparatus of claim 1, wherein the controller is further configured to: before determining the difference value between the first humidity and the second humidity,

- compare the first humidity and the second humidity to a predetermined target humidity; and based on both of the first humidity and the second humidity exceeding the predetermined target humidity, calculate the difference value between the first humidity and the second humidity.
6. The laundry treatment apparatus of claim 5, wherein the predetermined target humidity is a relative humidity of 60%.
7. The laundry treatment apparatus of claim 5, wherein the predetermined threshold value is a relative humidity of 10%.
8. The laundry treatment apparatus of claim 5, further comprising: a first actuator electrically connected to the controller and configured to operate the outlet flow path switching valve based on the control signal of the controller; and a second actuator electrically connected to the controller and configured to operate the inlet flow path switching valve based on the control signal of the controller.
9. The laundry treatment apparatus of claim 8, wherein the controller is further configured to: based on the first humidity being greater than the second humidity, produce a first control signal to increase the first opening degree of the first outlet and decrease the second opening degree of the second outlet according to a first predetermined ratio; transmit the first control signal to the first actuator; based on the second humidity being greater than the first humidity, produce a second control signal to increase the second opening degree of the second outlet and decrease the first opening degree of the first outlet according to a second predetermined ratio; and transmit the second control signal to the first actuator.
10. The laundry treatment apparatus of claim 9, wherein the controller is further configured to: based on the first humidity being greater than the second humidity, produce a third control signal to increase a third opening degree of the first inlet and decrease a fourth opening degree of the second inlet according to a third predetermined ratio; transmit the third control signal to the second actuator; based on the second humidity being greater than the first humidity, produce a fourth control signal to increase the fourth opening degree of the second inlet and decrease the third opening degree of the first inlet according to a fourth predetermined ratio; and transmit the fourth control signal to the second actuator.
11. A method for controlling a laundry treatment apparatus, the laundry treatment apparatus including a cabinet that defines an accommodation space therein, a first laundry treatment module disposed in a first portion of the accommodation space, a second laundry treatment module disposed in a second portion of the accommodation space, and a drying module configured to produce hot air and to supply the hot air to the first laundry treatment module and the second laundry treatment module through a first outlet and a second outlet, respectively, the drying module being further configured to receive return air from the first laundry treatment module and the second laundry treatment module through a first inlet and a second inlet, respectively, and to absorb at least a part of moisture included in the return air, the method comprising: supplying the hot air to the first and second laundry treatment modules through the first and second outlets, respectively; absorbing at least a part of the moisture included in the return air received through the first and second inlets from the first and second laundry treatment modules; measuring a first humidity of the return air received through the first inlet and a second humidity of the return air received through the second inlet; determining a difference value between the first humidity and the second humidity; determining whether the difference value exceeds a predetermined threshold value; and based on the difference value exceeding the predetermined threshold value, adjusting a ratio of amounts of the hot air to be supplied to the first laundry treatment module and the second laundry treatment module by changing a first opening degree of the first outlet or a second opening degree of the second outlet, or both.
12. The method of claim 11, wherein the first humidity is measured from the return air discharged from the first laundry treatment module by a first humidity sensor disposed adjacent to the first inlet; and wherein the second humidity is measured from the return air discharged from the second

laundry treatment module by a second humidity sensor disposed adjacent to the second inlet.

13. The method of claim 12, further comprising: comparing each of the first humidity and the second humidity to a predetermined target humidity; and determining whether each of the first humidity and the second humidity is less than or equal to the predetermined target humidity.

14. The method of claim 13, wherein the difference value is calculated based on both of the first humidity and the second humidity exceeding the predetermined target humidity.

15. The method of claim 13, wherein the predetermined target humidity is a relative humidity of 60%.

16. The method of claim 12, wherein the predetermined threshold value is a relative humidity of 10%.

17. The method of claim 13, wherein adjusting the ratio of the amounts of the hot air comprises: based on the first humidity being greater than the second humidity, increasing the first opening degree of the first outlet and decreasing the second opening degree of the second outlet according to a first predetermined ratio; and based on the second humidity being greater than the first humidity, increasing the second opening degree of the second outlet and decreasing the first opening degree of the first outlet according to a second predetermined ratio.

18. The method of claim 17, wherein adjusting the ratio of the amounts of the hot air further comprises: based on the first humidity being greater than the second humidity, increasing a third opening degree of the first inlet and decreasing a fourth opening degree of the second inlet according to a third predetermined ratio; and based on the second humidity being greater than the first humidity, increasing the fourth opening degree of the second inlet and decreasing the third opening degree of the first inlet according to a fourth predetermined ratio.

19. The method of claim 18, further comprising: supplying the hot air through the first and second outlets and absorbing at least a part of the moisture in the return air received through the first and second inlets after adjusting the ratio of the amounts of the hot air by changing at least one of the first opening degree of the first outlet, the second opening degree of the second outlet, the third opening degree of the first inlet, or the fourth opening degree of the second outlet.

20. The method of claim 19, further comprising: determining whether a predetermined time has elapsed since a start point of supplying the hot air through the first and second outlets after adjusting the ratio of the amounts of the hot air.

21. The method of claim 20, further comprising: based on determining that the predetermined time has elapsed since the start point, repeating operations that comprise: measuring the first humidity and the second humidity, comparing each of the first humidity and the second humidity to the predetermined target humidity, based on both of the first humidity and the second humidity exceeding the predetermined target humidity, calculating the difference value between the first humidity and the second humidity, and adjusting the ratio of the amounts of the hot air by changing at least one of the first opening degree of the first outlet, the second opening degree of the second outlet, the third opening degree of the first inlet, or the fourth opening degree of the second outlet.
